

Ashish Dhiman

MACHINE LEARNING ENGINEER

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13+ years

ML, research → production

US patent, granted

lane estimation from satellite imagery

2 research papers

EU-funded Earth observation

PROFILE

Lead Machine Learning Engineer with **13+ years** taking ML from research to production. Specialized in Representation Learning, Embeddings, and Computer Vision, with RAG and NLP. Proven at global scale across TomTom and HERE; inventor on a granted US patent and co-author of EU-funded research.

SKILLS

ML & Deep Learning

Representation Learning Embeddings Computer Vision NLP LLMs RAG Graph Neural Networks
Contrastive Learning Object Detection Segmentation MLOps

Frameworks

PyTorch PyTorch Lightning TensorFlow Keras scikit-learn pandas OpenCV

Platforms & Data

AWS Azure Spark Databricks Kubernetes Vector Databases Jenkins

Programming

Python Scala Java C# JavaScript Julia

Languages: English (Fluent), Hindi (Native)

EXPERIENCE

Lead Machine Learning Engineer — Applied Sciences

Feb 2020 – Present

TomTom · Gent, Belgium

- Architect and sole applied scientist for a leadership-sponsored moonshot: a unified **embedding layer** giving consistent vector representations across heterogeneous map data (roads, lanes, networks, POIs, signs) so internal and external teams build APIs, tools, and agents on one foundation. Own model selection, incremental delta-embedding generation, and hybrid deterministic + semantic retrieval; applying **Matryoshka Representation Learning** for adaptive multi-resolution embeddings. Build the system via spec-driven, agentic development — authoring specs and orchestrating AI coding agents.
- As internal ML consultant to the Guidance & Navigation org, built an end-to-end traffic-sign ↔ road map-matcher that raised matching accuracy from **67% to 99%**, plus a lane-restriction detection system and road-median → lane-border matching across navigation products.
- Led the HD Lane feature team — computer-vision models and map-matching over HD aerial and LiDAR data for autonomous-driving maps.
- Spearheaded an EU-co-funded research project ([Global Earth Monitor](#)) on automated global land-cover mapping; deep-learning classification and change detection cut global automation cost **40%** and produced two published papers.
- Built map-data representation models beyond the core platform: a **Graph Neural Network** road-network comparison system, and a contrastive-learning + vector-database system detecting suspicious OpenStreetMap changesets (anti-vandalism).
- Created a context-driven feature-prioritization system for TomTom Core map that cut data size while preserving quality, contributing to **five major client contracts**; won the TomTom Global Hackathon (2022).

Senior Data Scientist — HD Mapping

May 2016 – Jan 2020

HERE Maps · Mumbai, India

- Built and trained a **U-Net** model detecting road boundaries and lane markings from HD aerial imagery — core components of the company's ADAS mapping product.
- Built a satellite-imagery road-detection system (**EfficientNet**) generating road boundaries and centerlines, and a **YOLOv3** pipeline for painted road signs across North America.
- Automated geometry snapping (places → roads, one week → one day) and intersection lane-traversal detection (**~95%** manual time); won company hackathons three years running (2017–2019).

Founding Member, Tech Team

Oct 2017 – Aug 2018

Unfound · Mumbai, India

- Founding engineer; led a 5-person team building an NLP news-summarization platform, launched to **500+ active users** in the first quarter.

Senior Software Engineer

Jun 2012 – Apr 2016

Atos Worldline · Pune, India

- Built real-time banking web services and a customer-facing application (AngularJS / Java) deployed across multiple banking clients.

PROJECTS

- **Iron Log** — iOS strength-training app (Flutter, Firebase, SQLite), built solo via spec-driven, agentic development — authored PRDs/specs and orchestrated AI coding agents to implement and review; in App Store launch preparation.

EDUCATION

MS, Computer Science (Machine Learning Specialisation) — Georgia Institute of Technology, Atlanta

2020 – 2023

B.E., Electrical & Electronics — NIT Hamirpur

2008 – 2012

PUBLICATIONS & PATENT

- **Patent (inventor)** — Lane geometry and direction estimation using temporal satellite imagery for known road geometries ([US 11,391,576 B2](#)).
- Two research papers (Map-Making Use Case; Land-Cover Continuous Monitoring), as contributor — [EU CORDIS project results](#).